

Alessio Maiola

SOFTWARE ENGINEER

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Work Experience

Leonardo

CYBER TECHNOLOGY RESEARCHER

Rome, Italy

Jun. 2026 - Current

- Enrolled as a Cyber Technology Researcher to focus on advanced security architectures and threat mitigation strategies.

Daikin Applied Europe

SOFTWARE ENGINEER

Rome, Italy

Sep. 2025 - Jun. 2026

- Led the go-live and maintenance of a production-grade testing system, building full-stack **Python** and **Angular** automation features.
- Completely redesigned the **MSSQL** database to improve scalability and extensibility, inventing a custom tracking algorithm that reduced duplicate occurrences by over **90%**.
- Managed the technical scoping and vendor partnership for an externally-developed **RAG-based AI Agent**, driving the project from requirements collection to the initial vendor offer.
- Engineered **computer vision** pipelines to **100% automate** video processing and camera interaction for industrial testing.
- Executed a **CRA-aligned security assessment** to identify system vulnerabilities and implement architectural mitigation strategies.
- Deployed a centralized authentication system with **Microsoft Entra ID** to enable secure SSO and streamline user access.
- Automated CI/CD on-premise deployments using **GitHub Actions, NSSM, and Nginx**, cutting deployment overhead by **86%**.
- Leveraged LLMs to **reverse-engineer** proprietary communication protocols, developing a custom client to interface with closed-source PLCs.

Education

Sapienza Università di Roma

M.SC. IN ENGINEERING IN COMPUTER SCIENCE

Rome, Italy

Sep. 2023 - Oct. 2025

Final Grade 110 Cum Laude, **Thesis** An AI Framework for Linear B Translation into Ancient Greek and English, **GPA** 29.86/30

Sapienza Università di Roma

B.SC. IN INGEGNERIA INFORMATICA E AUTOMATICA

Rome, Italy

Sep. 2020 - Jul. 2023

Final Grade 110 Cum Laude, **Thesis** Change Detection su Immagini Satellitari mediante Reti Neurali, **Ranked 1st** in the faculty (**GPA** 29.90/30)

Main Projects

An AI Framework for Linear B Translation into Ancient Greek and English (Master Thesis)

Available on Github

ARTIFICIAL INTELLIGENCE AND INTERACTIVE GRAPHICS LABORATORY PROJECT

2025

I built an end-to-end **AI pipeline** to translate Mycenaean Linear B into Ancient Greek, and then into English. The system integrates a **state-of-the-art cognate matching model** with targeted **prompt-engineering** strategies, and custom tasks designed to improve translation accuracy.

GROSSO - Automatic Prompter for Attack/Defense CTFs

Available on Github

PERSONAL PROJECT

2025

Enables **API-based** communication with an informed instance of **Gemini** for **automatic vulnerability detection** and **exploit generation**, featuring an interactive chat interface for **exploit refinement**.

EMOFY - Full Stack Application Development

Available on Github

SOFTWARE ENGINEERING PROJECT

2024

An **AI-powered distributed application** leveraging **Docker** and **Kubernetes** to automate labeled dataset creation.

Video Prediction - Computer Vision

Available on Github

COMPUTER VISION PROJECT

2024

Testing and evolving **state-of-the-art spatiotemporal attention models** for video prediction on MovingMNIST and UCF101 datasets.

Skills

Full-Stack	Python, Java, Go, C, TypeScript, Spring Boot, FastAPI, Node.js, Next.js, Angular, React
DevOps	Microservices, Docker, Kubernetes, GitHub Actions, Jenkins, Nginx
Data Modeling & Databases	MSSQL, PostgreSQL, MySQL, NoSQL (document, column, graph), Query Optimization
AI & Machine Learning	Deep Learning, Transformers, Computer Vision, NLP, Reinforcement Learning, Pytorch, CrewAI, LangChain
Cybersecurity	Binary Exploitation, Cryptography, Web Security, Reverse Engineering, Network Security

Honors & Awards

2025	Winner of the National Attack/Defense CTF Competition – Team Sapienza Captain , CyberChallenge.IT	Turin, Italy
2025	Best Presentation Award – National A/D CTF Competition , CyberChallenge.IT	Turin, Italy
2025	Top 5% in the National Jeopardy CTF Competition , CyberChallenge.IT	Rome, Italy
2025	Selected for the CyberChallenge program at Sapienza (Top 5% of 355 participants) , CyberChallenge.IT	Rome, Italy
2019	Awarded for excellent results in Mathematics Olympiad individual competition , Accademia dei Lincei	Rome, Italy